A Unified Transmission Strategy for TDD/FDD Massive MIMO Systems With Spatial Basis Expansion Model

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***Abstract*—This paper proposes a unified transmission strategy for multiuser time division duplex (TDD)/frequency division duplex (FDD) massive multiple-input–multiple-output (MIMO) systems, including uplink (UL)/downlink (DL) channel estimation and user scheduling for data transmission. With the aid of antenna array theory and array signal processing, we build a spatial basis expansion model (SBEM) to represent the UL/DL channels with far fewer parameter dimensions. Hence, both the UL and DL channel estimations of multiusers can be carried out with a small amount of training resource, which significantly reduces the training overhead and feedback cost. Meanwhile, the pilot contamination problem in the UL training is immediately relieved by exploiting the spatial information of users. To enhance the spectral efficiency, we also design a greedy user scheduling scheme during the data transmission period. Compared with existing low-rank models, the newly proposed SBEM offers an alternative for channel acquisition without the need for channel statistics and can be applied to both TDD and FDD systems. Various numerical results are provided to corroborate the proposed studies.**

***Index Terms*—Angle reciprocity, direction of arrival (DOA), discrete Fourier transform (DFT), massive multiple input multiple output (MIMO), spatial basis expansion model (SBEM).**

I. Introduction

LARGE-SCALE multiple-input–multiple-output (MIMO) or "massive MIMO" systems [1] have drawn considerable interest from both academia and industry. Theoretically, massive MIMO systems can almost perfectly relieve the inter-user interference in multiuser MIMO systems with simple linear transceivers [2]. It was shown in [3] that each antenna element of a very large MIMO system consumes exceedingly low power, and the total power can be made inversely proportional to the number of antennas. Other advantages, such as high spectral efficiency, security, robustness, or reliable linkage, also play key roles in promoting massive MIMO systems appealing for the next generation of wireless systems [3], [4].

However, all these potential gains rely heavily on the perfect channel state information (CSI) at the base station (BS). From the conventional orthogonal training strategy [5], the optimal number of training streams should be the same as the number of transmit antennas, and the length of the training should be no less than the number of transmit antennas. Hence, downlink (DL) training in a massive MIMO system requires huge number of orthogonal training sequences. This severe overhead and the accompanied high calculation complexity will overwhelm the system performance and mitigate any possible improvement. For uplink (UL) training, the conventional channel estimation methods are generally feasible. However, as the number of users or the number of user's antennas grows, the increased pilot overhead will deteriorate the system efficiency and become the system bottleneck. If the nonorthogonal sequences are used, then the so-called pilot contamination will also deteriorate the system performance.

For time division duplex (TDD) massive MIMO systems, DL CSI can be obtained by leveraging the channel reciprocity [6], which has promoted quite many research studies [7]–[11]. However, in practice, the channel reciprocity between UL and DL may not exactly hold even for TDD systems due to the calibration error between the UL/DL radio frequency chains [12]. In addition, the property of channel reciprocity has been proven to be robust only for the single-cell scenario [13], and this will undoubtedly increase the pressure of multicell coordination. On the other hand, the DL channel estimation for frequency division duplex (FDD) massive MIMO systems is always deemed as a difficult problem since the channel reciprocity does not hold and cannot be used to simplify the estimation. In [2], [14], and [15], the closed-loop training schemes were applied to sequentially design the optimal pilot beam patterns. The compressive sensing (CS)-based feedback reduction in [16] and the distributed compressive channel estimation in [17] exploited the channel sparsity to reduce the heavy burden of feeding back large amount of measurements.

Except for the aforementioned regular attempts, a new way to design the transmission strategy for massive MIMO systems is to exploit the low-rank approximation of channel covariance matrix [18]–[20]. Based on the assumption that the angular spread (AS) of the incident signals at the BS from each user is narrow, in [18]–[20], it was proposed to reduce the dimensionality of the effective channels through eigendecomposition

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of channel covariance matrices. All these covariance-aware methods could be categorized into *spatial division multiplexing*, which utilizes nonoverlapped spatial information of different users to realize the orthogonal transmission. Meanwhile, compressed channel sensing has been widely adopted for channel sparsity models, such as the Karhunen–Loeve transform basis representation in [17] and the virtual angle representation in [21]. As the channel sparsity patterns in these existing methods are often assumed unknown, nonlinear reconstruction procedures are thus inevitable.

In this paper, we build an alternative low-rank model for massive uniform linear array (ULA) from antenna array theory and array signal processing viewpoint. The new model relies on the mean direction of arrivals (DOA) and the AS of the incident signals of each user and is named spatial basis expansion model (SBEM). The proposed SBEM can be viewed as *modified* virtual channel representation (VCR) [22] but could represent the channel in a much sparser format. We then present a unified transmission strategy for the multiuser TDD/FDD massive MIMO systems, including UL/DL channel estimation and user scheduling for data transmission. It is shown that the UL/DL channel estimation of multiusers can be carried out with very few training resources, and thus, the overhead of training and feedback can be also reduced significantly. Meanwhile, the pilot contamination in UL training can be immediately relieved. To enhance the spectral efficiency during the data transmission, we also propose a greedy user scheduling algorithm where users with orthogonal spatial information are allowed to transmit simultaneously. Compared with the existing low-rank models [17]–[22], the proposed one offers the following several benefits.

1) SBEM is designed from array antenna theory and can offer part of channel reciprocity for both TDD/FDD massive MIMO systems. By contrast, FDD reciprocity has not been reported either from covariance-based methods [18], [20] or CS-based methods [17], [21].
2) SBEM exploits the structure of antenna arrays and offers a sparser channel representation than those in [17] and [21], with a spatial rotation operation.
3) DOA and AS for SBEM could be simply tracked from users' data in a blind manner.
4) SBEM does not need the knowledge of either the UL or the DL channel covariances.
5) SBEM could be simply implemented by fast Fourier transform (FFT) and partial FFT [23]–[25], followed by a linear estimator. By contrast, covariance-based methods [18], [20] need eigenvalue decomposition (EVD) for high-dimensional covariance matrices, and CS-based methods [17], [21] require nonlinear optimization.[1]

The rest of this paper is organized as follows. In Section II, the system model and the channel characteristics with narrow AS are described. The SBEM-aided channel estimations for UL/DL transmission are presented in Section III. Section IV designs the user scheduling for data transmission, followed by simulations in Section V. Finally, conclusions are drawn in Section VI.

Notations: Vectors and matrices are denoted by boldface small and capital letters; the transpose, complex conjugate, Hermitian, inverse, and pseudoinverse of the matrix $\mathbf{A}$ are denoted by $\mathbf{A}^T$, $\mathbf{A}^*$, $\mathbf{A}^H$, $\mathbf{A}^{-1}$, and $\mathbf{A}^\dagger$, respectively; $\mathrm{tr}(\mathbf{A})$ is the trace of $\mathbf{A}$; $[\mathbf{A}]_{ij}$ is the (i,j)th entry of $\mathbf{A}$; the entry index of the vector and the matrix starts from 0; $\mathrm{diag}\{\mathbf{a}\}$ denotes a diagonal matrix with the diagonal elements constructed from $\mathbf{a}$, and $\mathrm{diag}\{\mathbf{A}\}$ denotes a vector whose elements are extracted from the diagonal components of $\mathbf{A}$; $\mathbf{I}$ is the identity matrix with appropriate size, and $\mathbb{E}\{\cdot\}$ is the statistical expectation; $\lceil x \rceil$ denotes the smallest integer no less than x, $\lfloor x \rfloor$ denotes the largest integer no more than x, and $\lfloor x \rceil$ denotes the integer closest to x; $[\mathbf{H}]_{:,\mathcal{D}}$ denotes the submatrix of $\mathbf{H}$ by collecting the columns indexed by $\mathcal{D}$, and $[\mathbf{H}]_{\mathcal{D},:}$ denotes the submatrix of $\mathbf{H}$ by collecting the rows indexed by $\mathcal{D}$; $[\mathbf{h}]_{\mathcal{D},:}$ indicates the subvector of $\mathbf{h}$ by keeping the elements indexed by $\mathcal{D}$; $|\mathcal{D}|$ denotes the cardinality of the set $\mathcal{D}$, and $\|\mathbf{h}\|$ denotes the Euclidean norm of $\mathbf{h}$; "\" defines the set subtraction operation, and $\delta(\cdot)$ is the Dirac delta function.

II. System Model and Channel Characteristics

A. System Model

Array-structure-based physical channel models have been widely adopted for MIMO systems, such as the spatial channel model [26], which exploits the array manifold and information of DOA and the direction of departures (DOD) of propagation signals. For massive MIMO systems, the significantly improved spatial resolution of large-scale antenna arrays and plenty of emerging low-complexity DOA acquisition techniques [27], [28] have further promoted these physical channel models [18]–[20]. Many studies [12], [19], [29] of massive MIMO systems require the BS to be elevated at a very high altitude, for example, on the top of a high building or a dedicated tower, such that there are few surrounding scatterers at the end of the BS. In this case, the incident AS seen by the BS array is usually limited in a narrow region [18]–[20].

Let us consider a multiuser massive MIMO system, where the BS is equipped with $M \gg 1$ antennas in the form of ULA, and K single-antenna users are randomly distributed in the coverage area. For a better illustration of our key idea, we assume that the channel is flat fading for the time being as done in [18] and [20], while the proposed idea can be easily extended to the frequency-selective scenario.

The propagation from user k to the BS is assumed to be composed of $P(P \gg 1)$ rays, and the corresponding $M \times 1$ UL channel can be expressed as [18], [20]

$$\mathbf{h}_k = \frac{1}{\sqrt{P}} \sum_{p=1}^{P} \alpha_{kp} \mathbf{a}(\theta_{kp}) \tag{1}$$

where α_{kp} represents the complex gain of the pth ray. Moreover, $\mathbf{a}(\theta_{kp}) \in \mathbb{C}^{M\times 1}$ is the array manifold vector and has the form

$$\mathbf{a}(\theta_{kp}) = \left[1, e^{j\frac{2\pi d}{\lambda}\sin\theta_{kp}}, \ldots, e^{j\frac{2\pi d}{\lambda}(M-1)\sin\theta_{kp}}\right]^T \tag{2}$$

[1]Although there exists several low-complexity CS algorithms, such as orthogonal matched pursuit, most of them still rely on nonlinear optimization.

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TABLE I
EXAMPLE OF OFFLINE TABLE OF C_k, WHERE $M = 128$, $\eta = 0.95$, AND $\theta_k \in [1^\circ,\ 90^\circ]$

θ_k	1°	3°	5°	7°	9°	11°	13°	15°	17°	19°	21°	23°	25°	27°	29°
C_k	1	8	9	4	1	5	9	9	7	4	1	1	1	1	1
θ_k	31°	33°	35°	37°	39°	41°	43°	45°	47°	49°	51°	53°	55°	57°	59°
C_k	1	3	7	10	6	1	8	6	4	7	6	1	9	7	3
θ_k	61°	63°	65°	67°	69°	71°	73°	75°	77°	79°	81°	83°	85°	87°	89°
C_k	1	1	1	1	6	10	4	4	8	4	5	9	5	1	1

where d is the antenna spacing, λ denotes the signal carrier wavelength, and θ_{kp} represents the DOA of the pth ray.

Similar to [18] and [20], we assume that the incident angles of user k with mean DOA θ_k are limited in a narrow angular range $[\theta_k - \Delta\theta_k, \theta_k + \Delta\theta_k]$, where $\Delta\theta_k$ is the single-side AS. Within this narrow angular range, it is obvious that $\mathbf{a}(\theta_{kp})$ are highly correlated with each other such that the channel covariance matrix of user k, which is defined as $\mathbf{R}_{h_k} = E\{\mathbf{h}_k\mathbf{h}_k^H\}$, approximately possesses the low-rank property.

In this paper, we present a low-rank property from antenna array theory, targeting at further reducing the channel estimation complexity. Due to the high correlations among $\mathbf{a}(\theta_{kp})$, $p = 1, \ldots, P$, $\mathbf{h}_k$ could be expanded from $\nu \ll P$ limited orthogonal basis as

$$\mathbf{h}_k = \sum_{q=1}^{\nu} \kappa_q \mathbf{b}_q, \quad \text{for } k = 1, \ldots, K \tag{3}$$

where $\mathbf{b}_q$ are the basis vectors to be determined, and κ_q are the corresponding coefficients. If $\nu \ll M$ also holds, then (3) is also known as the basis expansion model (BEM) [30]. Hence, estimating M unknown elements in $\mathbf{h}_k$ will be greatly simplified to estimating ν constants κ_q only.

B. Characteristics of ULA and Channel Vectors

Define the normalized discrete Fourier transform (DFT) of the channel vector $\mathbf{h}_k$ as $\tilde{\mathbf{h}}_k = \mathbf{F}\mathbf{h}_k$, where $\mathbf{F}$ is the $M \times M$ DFT matrix whose (p,q)th element is $[\mathbf{F}]_{pq} = e^{-j(2\pi/M)pq}/\sqrt{M}$. As per antenna array theory, $\tilde{\mathbf{h}}_k$ will exhibit some specific properties in massive ULA case that could be further exploited to achieve low-complexity transceiver design.

Lemma 1: For the simplest one-ray case, i.e., $\mathbf{h}_k = \alpha_k\mathbf{a}(\theta_k)$, $\tilde{\mathbf{h}}_k$ has only one nonzero point as $M \to \infty$, and this nonzero point reflects the DOA of the impinging signal. When M is large but not infinite, most power of $\tilde{\mathbf{h}}_k$ still concentrates around this nonzero point.

Proof: The qth component of $\tilde{\mathbf{h}}_k$ is computed as

$$\begin{aligned}[\tilde{\mathbf{h}}_k]_q &= \frac{\alpha_k}{\sqrt{M}} \sum_{m=0}^{M-1} e^{-j\left(\frac{2\pi}{M}mq - \frac{2\pi}{\lambda}md\sin\theta_k\right)} \\ &= \frac{\alpha_k}{\sqrt{M}} e^{-j\frac{M-1}{2}\left[\frac{2\pi}{M}q - \frac{2\pi}{\lambda}d\sin\theta_k\right]} \\ &\quad \cdot \frac{\sin\left[\left(\frac{2\pi}{M}q - \frac{2\pi}{\lambda}d\sin\theta_k\right)\frac{M}{2}\right]}{\sin\left[\left(\frac{2\pi}{M}q - \frac{2\pi}{\lambda}d\sin\theta_k\right)\frac{1}{2}\right]}. \end{aligned} \tag{4}$$

When $M \to \infty$, it is obvious that

$$\lim_{M\to\infty} \left|[\tilde{\mathbf{h}}_k]_q\right| = |\alpha_k| \cdot \sqrt{M}\delta\left(\frac{q}{M} - \frac{d\sin\theta_k}{\lambda}\right) \tag{5}$$

which means that there always exists a q_0 that satisfies $q_0 = M(d/\lambda)\sin\theta_k$ for any possible θ_k. This nonzero point q_0, in turn, reflects the DOA of the incident signal, namely, $\theta_k = \arcsin(q_0\lambda/Md)$. On the other hand, when M is large but not infinite, the power leakage may happen. Specifically, for most other cases, $M(d/\lambda)\sin\theta_k$ is not an integer, and the channel power will leak from the $(\lfloor M(d/\lambda)\sin\theta_k \rceil)$th DFT point to other DFT points. In fact, DFT outputs are discrete samples of the discrete-time Fourier transform (DTFT) of $\mathbf{a}(\theta_k)$, i.e., a Sinc function, at the points of $2\pi q/M$, $q = 0, \ldots, M-1$. Hence, the degree of leakage in $\tilde{\mathbf{h}}_k$ is inversely proportional to M. When M is large but not infinite, the power leakage is not serious and most power of $\tilde{\mathbf{h}}_k$ will still concentrate around $M(d/\lambda)\sin\theta_k$. ■

From *Lemma 1*, it is clear that most channel power concentrates on DFT points in the vicinity of $\lfloor M(d/\lambda)\sin\theta_k \rceil$ when M is large but not infinite. It is then of interest to find the minimal number of DFT points around $\lfloor M(d/\lambda)\sin\theta_k \rceil$ that contain at least η percentage of the total power. Define the corresponding set of DFT points as $\mathcal{C}_k$, which can be found from the following optimization:

$$\begin{aligned} \min_{\mathcal{C}_k} \quad & C_k \triangleq |\mathcal{C}_k| \\ \text{s.t.} \quad & \sum_{q\in\mathcal{C}_k} \frac{1}{M}\left\{\frac{\sin\left[\left(\frac{2\pi}{M}q - \frac{2\pi}{\lambda}d\sin\theta_k\right)\frac{M}{2}\right]}{\sin\left[\left(\frac{2\pi}{M}q - \frac{2\pi}{\lambda}d\sin\theta_k\right)\frac{1}{2}\right]}\right\}^2 \geq \eta. \end{aligned} \tag{6}$$

The optimization in (6) is difficult to solve in closed form. Nevertheless, since (6) is merely dependent on the array configuration, we can establish the offline tables for $\theta_k \in (-(\pi/2), \pi/2)$ and some predefined η. An example of the offline table of C_k for a single incident ray is given in Table I, with $M = 128$ and $\eta = 95\%$, where C_k is a function of θ_k and η. It is shown in Table I that the maximal number of leakage points that contain at least 95% of the channel power is 10 for any incident angle, which is very small compared to the total antenna number $M = 128$.

Lemma 2: For the multiray channel (1) with narrow AS, $\tilde{\mathbf{h}}_k$ only has nonzero elements at limited points as $M \to \infty$, each corresponding to a single ray of DOA within $[\theta_k - \Delta\theta_k, \theta_k + \Delta\theta_k]$.

Proof: Based on (5) and considering the multiray channel, i.e., $\mathbf{h}_k = (1/\sqrt{P})\sum_{p=1}^{P}\alpha_k\mathbf{a}(\theta_{kp})$, it can be similarly

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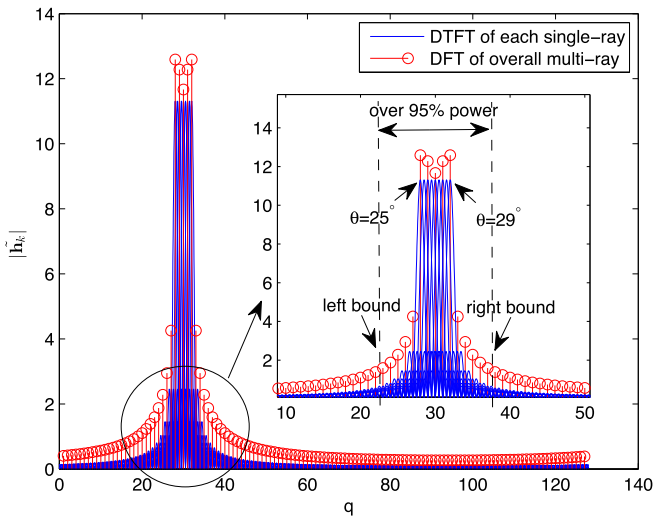


Fig. 1. Example of a multiray channel with incident angles in [25°, 29°]. $M = 128$, and $d = \lambda/2$.

found that each of the P rays will be mapped to a single DFT point as $M \to \infty$, and then the index set of nonzero DFT points of $\tilde{\mathbf{h}}_k$ can be determined as

$$\mathcal{D}_k = \left\{ q \left| M\frac{d}{\lambda}(\sin\theta_{kp})_{\min} \le q \le M\frac{d}{\lambda}(\sin\theta_{kp})_{\max} \right. \right. \\ \left. \left. \forall\, \theta_{kp} \in [\theta_k - \Delta\theta_k, \theta_k + \Delta\theta_k] \right\},\ M \to \infty. \quad (7)$$

When M is large but not infinite, let us redefine $\mathcal{D}_k$ as the index set of the continuous DFT points that contain at least η percentage of the total channel power. Then, the upper bound of the cardinality of $\mathcal{D}_k$ can be approximated as

$$\begin{aligned} |\mathcal{D}_k| &\le M\frac{d}{\lambda}(\sin\theta_{kp})_{\max} - M\frac{d}{\lambda}(\sin\theta_{kp})_{\min} + C_{\max} \\ &\le \left\lceil M\frac{d}{\lambda}\sin(\theta_k+\Delta\theta_k)\right\rceil - \left\lfloor M\frac{d}{\lambda}\sin(\theta_k-\Delta\theta_k)\right\rfloor + 1 + C_{\max} \\ &\le \left\lceil 2M\frac{d}{\lambda}\cdot|\cos\theta_k|\cdot\Delta\theta_k + 1\right\rceil + C_{\max} \end{aligned} \quad (8)$$

where $C_{\max}$ is the possible maximum number of leakage and is defined as $C_{\max} \triangleq \max\ C_k$ for a given η and all $\theta \in [\theta_k - \Delta\theta_k, \theta_k + \Delta\theta_k]$.

To illustrate this, an example of a nine-ray channel with incident angles in [25°, 29°] is given in Fig. 1, where the DTFT of each single ray and the DFT of overall multirays are depicted, respectively. Obviously, the actual cardinality of $\mathcal{D}_k$ containing $\eta = 95\%$ power is only 15, which is quite small compared to the total antenna number $M = 128$. ■

Following *Lemma 2* and bearing in mind that $\Delta\theta_k$ is small, it is readily known that $|\mathcal{D}_k| \le \lceil (2Md/\lambda)\cdot|\cos\theta_k|\cdot\Delta\theta_k + 1\rceil + C_{\max}$ is still small compared to M. Hence, $\tilde{\mathbf{h}}_k$ also has its most power concentrated in limited number of entries.

Corollary 1: The index set $\mathcal{D}_k$ of $\tilde{\mathbf{h}}_k$ reflects both the mean DOA and the AS of the incoming signals.

Therefore, the channel vector of massive MIMO can be approximated with fewer parameters as

$$\mathbf{h}_k = \mathbf{F}^H\tilde{\mathbf{h}}_k \approx [\mathbf{F}^H]_{:,\mathcal{D}_k}[\tilde{\mathbf{h}}_k]_{\mathcal{D}_k,:} = \sum_{q\in\mathcal{D}_k}\tilde{h}_{k,q}\mathbf{f}_q \quad (9)$$

where $\tilde{h}_{k,q} \triangleq [\tilde{\mathbf{h}}_k]_q$ denotes the qth element of $\tilde{\mathbf{h}}_k$, and $\mathbf{f}_q$ is the qth column of $\mathbf{F}^H$. Comparing with (3), (9) is exactly in the form of a BEM with orthogonal basis vectors $\mathbf{f}_q$.

Interestingly, the DFT vector $\mathbf{f}_q$ coincides with the steering vector as $\mathbf{f}_q = \mathbf{a}(\theta_q)$, where $\theta_q = \arcsin(q\lambda/Md)$, which means that $\mathbf{f}_q$ formulates an array beam toward the physical direction $\theta_q = \arcsin(q\lambda/Md)$. Hence, all beams $\mathbf{f}_q$, $q \in \mathcal{D}_k$ will point toward the AS of user k and are orthogonal to each other. Consequently, the beam indices $\mathcal{D}_k$ can be viewed as the *spatial signature* of user k [31], and (9) can be deemed as the SBEM, as named after the popular temporal BEM [30].

Note that (9) coincides with the conventional VCR [22] but is particularly deduced here for the massive antennas systems. Meanwhile, the conventional VCR could be further modified from the perspective of antenna array theory. Specifically, it is expressed as follows.

Definition 1: Define

$$\tilde{\mathbf{h}}_k^{\mathrm{ro}} = \mathbf{F}\mathbf{\Phi}(\phi)\mathbf{h}_k \quad (10)$$

as the *modified VCR*, where

$$\mathbf{\Phi}(\phi) = \mathrm{diag}\left\{\left[1, e^{j\phi}, \ldots, e^{j(M-1)\phi}\right]\right\} \quad (11)$$

with $\phi \in [-(\pi/M), \pi/M]$ being a *spatial rotation parameter*.[2]

The *spatial rotation* operation $\tilde{\mathbf{h}}_k^{\mathrm{ro}} = \mathbf{F}\mathbf{\Phi}(\phi)\mathbf{h}_k$ can further concentrate channel power within fewer entries of $\tilde{\mathbf{h}}_k^{\mathrm{ro}}$ for a certain value of ϕ. For example, for the one-ray case, when the DOA of the incident signal is not $\arcsin(q_0\lambda/Md)$ for some integer q_0, i.e., mismatched with DFT points, then power leakage will happen. Formulate a new channel vector as $\tilde{\mathbf{h}}_k^{\mathrm{ro}} = \mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{h}_k$. If $\phi_k = ((2\pi q_0/M) - (2\pi d/\lambda)\sin\theta_k)$, then $\tilde{\mathbf{h}}_k^{\mathrm{ro}}$ has only one nonzero element $\tilde{h}_{k,q_0}^{\mathrm{ro}}$ at q_0, with the power leakage eliminated.

On the other hand, for the multiray case, power leakage always happens, but spatial rotation can still help to gather more channel power on fewer DFT points, namely, to mitigate the leakage and to strengthen the channel sparsity. Specifically, we can also formulate $\tilde{\mathbf{h}}_k^{\mathrm{ro}} = \mathbf{F}\mathbf{\Phi}(\phi)\mathbf{h}_k$ for the multiray channel and define $\mathcal{D}_k^{\mathrm{ro}}$ as the continuous index set such that $[\tilde{\mathbf{h}}_k^{\mathrm{ro}}]_{\mathcal{D}_k^{\mathrm{ro}},:}$ contains at least η percentage of the channel power. Next, we may search ϕ over the small range $[-(\pi/M), \pi/M]$ and select the optimal ϕ_k that minimizes $|\mathcal{D}_k^{\mathrm{ro}}|$.

An example of a multiray channel with $\theta_k = 60°$ and $\Delta\theta_k = 2°$ is given in Fig. 2, where $M = 128$, $d = \lambda/2$, and $\eta = 0.99$. It can be seen that the cardinality of $\mathcal{D}_k^{\mathrm{ro}}$ is only $|\mathcal{D}_k^{\mathrm{ro}}| = 11$ after spatial rotation, whereas the cardinality before the rotation is $|\mathcal{D}_k| = 28$. Furthermore, the averaged curves of $|\mathcal{D}_k|$ and $|\mathcal{D}_k^{\mathrm{ro}}|$ as a function of AS have been depicted in Fig. 3. It can be seen

[2]The possible range of ϕ is set as $[-(\pi/M), \pi/M]$ for the spatial resolution of M-point DFT is exactly $2\pi/M$.

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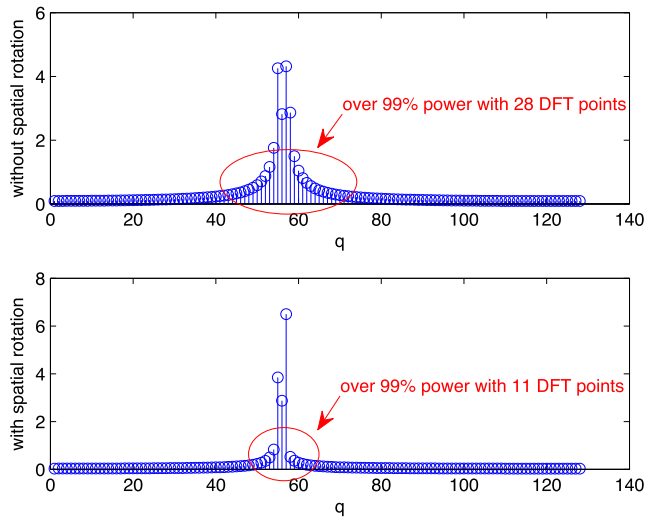


Fig. 2. Comparison of multiray channels with/without spatial rotation, where $\theta_k = 60°$, $\Delta\theta_k = 2°$, $M = 128$, $d = \lambda/2$, and $\eta = 0.99$.

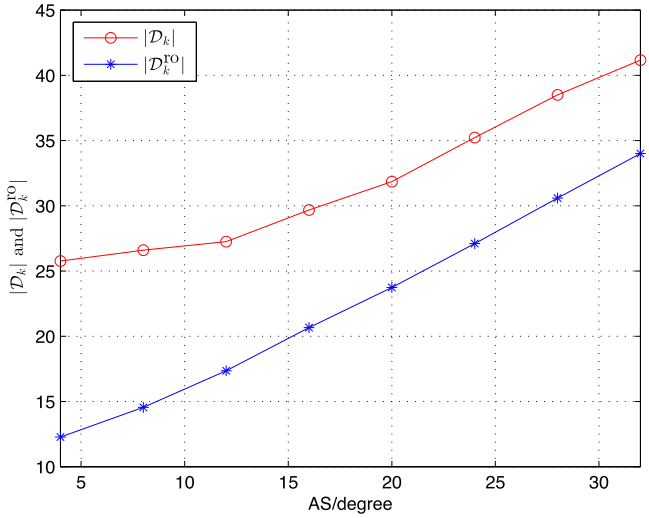


Fig. 3. Comparison of $|\mathcal{D}_k|$ and $|\mathcal{D}_k^{\mathrm{ro}}|$ for multiray channels without/with spatial rotation, where $M = 128$, $d = \lambda/2$, and $\eta = 0.99$.

that, as the AS varies, there is always an obvious reduction from $|\mathcal{D}_k|$ to $|\mathcal{D}_k^{\mathrm{ro}}|$. This means that the spatial rotation can help to strengthen the channel sparsity for even larger AS.

Therefore, we can rerepresent $\mathbf{h}_k$ as

$$\mathbf{h}_k = \mathbf{\Phi}(\phi_k)^H \mathbf{F}^H \tilde{\mathbf{h}}_k^{\mathrm{ro}} \approx \mathbf{\Phi}(\phi_k)^H [\mathbf{F}^H]_{:,\mathcal{D}_k^{\mathrm{ro}}} \left[\tilde{\mathbf{h}}_k^{\mathrm{ro}}\right]_{\mathcal{D}_k^{\mathrm{ro}},:}$$
$$= \sum_{q \in \mathcal{D}_k^{\mathrm{ro}}} \tilde{h}_{k,q}^{\mathrm{ro}} \mathbf{\Phi}(\phi_k)^H \mathbf{f}_q \tag{12}$$

where $\mathbf{\Phi}(\phi_k)^H \mathbf{f}_q$, $q \in \mathcal{D}_k^{\mathrm{ro}}$, are the new equivalent basis vectors. Since these new basis vectors are still mutually orthogonal, (12) is also a type of SBEM.

Corollary 2: The operation $\mathbf{\Phi}(\phi_k)\mathbf{h}_k$ can be viewed as rotating the orthogonal beams $\mathbf{f}_q$ simultaneously such that the new beams $\mathbf{\Phi}(\phi_k)^H \mathbf{f}_q$ point toward user k more accurately.

Remark 1: The proposed SBEM is different from the conventional sparse models [17], [21] in that the position $\mathcal{D}_k^{\mathrm{ro}}$ can be obtained from angular parameter estimation. Hence, the linear estimator can be applied to estimate the remaining parameters $\tilde{h}_{k,q}^{\mathrm{ro}}$. On the other hand, in [17] and [21], the positions of nonzero points are considered as unknown during channel estimation, and hence, the nonlinear optimization has to be applied. Moreover, without harnessing the antenna array theory to enhance the channel sparsity, these CS-based methods will have a relatively larger sparsity level, resulting in lower channel estimation accuracy, particularly when the channel power leakage is severe.

C. Practical Consideration

For practical applications, the standards normally regulate a fixed number of channel parameters to be estimated instead of considering a dynamic number $|\mathcal{D}_k^{\mathrm{ro}}|$. Denote τ as the number of the channel parameters that the system could handle, i.e., the system could support at most τ orthogonal training sequences. Define the set containing continuous τ integers as $\mathcal{B}_k^{\mathrm{ro}}$. Then, we should select the spatial signature set $\mathcal{B}_k^{\mathrm{ro}}$ and the corresponding rotation parameter ϕ_k for user k according to

$$\max_{\phi_k,\, \mathcal{B}_k^{\mathrm{ro}}} \left\| \left[\tilde{\mathbf{h}}_k^{\mathrm{ro}}\right]_{\mathcal{B}_k^{\mathrm{ro}},:} \right\|^2 \quad \text{subject to } |\mathcal{B}_k^{\mathrm{ro}}| = \tau. \tag{13}$$

The preceding optimization can be achieved by adopting a 1-D search over $\phi \in [-(\pi/M), \pi/M]$ and for each possible ϕ, sliding a window of size τ over the elements in $\tilde{\mathbf{h}}_k^{\mathrm{ro}}$ to determine $\mathcal{B}_k^{\mathrm{ro}}$ that maximizes the optimization objective.

Remark 2: For massive MIMO, the search range of ϕ, i.e., $[-(\pi/M), \pi/M]$, is very small, and the accompanied complexity is low.

III. Unified Channel Estimation for Time Division Duplex/Frequency Division Duplex System

We propose a unified UL/DL transmission strategy for a TDD/FDD massive MIMO system that utilizes the spatial signatures to realize the orthogonal training and data transmission among different users. As shown in Fig. 4, the transmissions between the BS and the users always start from a UL preamble to obtain the spatial signature of each user. Then, users are grouped and scheduled for UL/DL training and data transmission based on their spatial signatures such that the orthogonal transmission is achieved. Based on the proposed SBEM, the channel parameters $\tilde{\mathbf{h}}_k^{\mathrm{ro}}$ can be easily estimated through traditional linear least square (LS) method. However, to make the proposed strategy complete, we will still present the whole channel estimation procedures.

A. Obtain Spatial Information Through UL Preamble

Assume that the current cell[3] is allocated with $\tau < K$ orthogonal training sequences[4] of length $L < T$, where T is

[3]In this paper, we focus on reducing the pilot overhead for each single cell such that the extra orthogonal training sequences can be allocated to the other cells. In this case, the pilot contamination from neighboring cells is totally avoided, and then our method can be directly extended to multicell scenarios.

[4]The number of orthogonal training sequences for each cell is set as the number of unknown channel parameters, i.e., $|\mathcal{B}_k^{\mathrm{ro}}| = \tau$ in (13), because this is exactly the minimum pilot overhead required.

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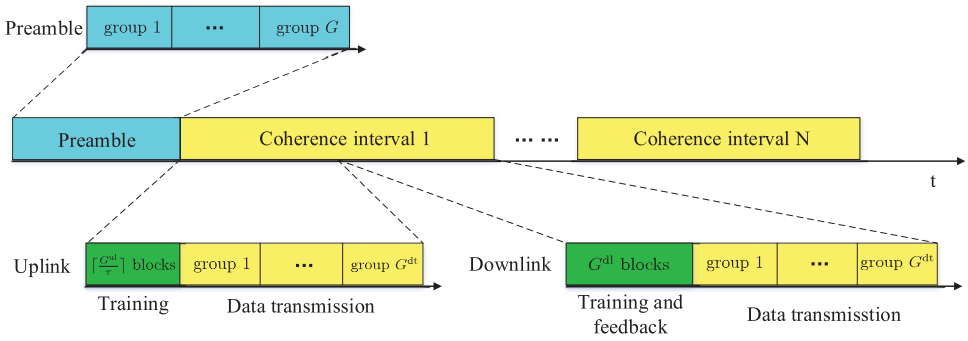


Fig. 4. Communication process of the new framework.

the channel coherence time, for both UL and DL training. Denote the corresponding orthogonal training set as $\mathcal{S}_{\rm cell} = \{\mathbf{s}_1, \ldots, \mathbf{s}_\tau\}$, with $\mathbf{s}_i^H \mathbf{s}_j = L\sigma_p^2 \cdot \delta(i-j)$, where σ_p^2 is the pilot signal training power. For ease of illustration, assume that $K = G\tau$, where $G > 1$ is an integer.

Following most standards, there exists a relatively long UL training period, which is called *preamble*, at the very beginning of any transmission, and the functionality of the preamble in the proposed framework is mainly to obtain the spatial signatures of different users. Since we do not assume any prior spatial information, we will have to divide K users into G groups, each containing τ users such that τ orthogonal training sequences are sufficient for each group. Then, the conventional UL channel estimation will be applied for each group during the preamble period, and hence, the length of the preamble is GL.

Taking the first group as an example, the received training signals at the BS are given by

$$\mathbf{Y} = \mathbf{H}\mathbf{D}^{\frac{1}{2}}\mathbf{S}^H + \mathbf{N} = \sum_{i=1}^{\tau} \sqrt{d_i}\mathbf{h}_i \mathbf{s}_i^H + \mathbf{N} \tag{14}$$

where $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_\tau] \in \mathbb{C}^{M\times\tau}$ is the UL channel matrix from the first τ users; $\mathbf{S} \triangleq [\mathbf{s}_1, \ldots, \mathbf{s}_\tau] \in \mathbb{C}^{L\times\tau}$ contains τ training sequences assigned to this cell; $\mathbf{D} \triangleq \mathrm{diag}\{d_1, \ldots, d_\tau\}$ and $d_k \triangleq P_k^{\rm ut}/L\sigma_p^2$ are used to satisfy the UL training energy constraint $P_k^{\rm ut}$ of user k; and $\mathbf{N}$ is the independent additive white Gaussian noise matrix with elements distributed as independent and identically distributed $\mathcal{CN}(0, \sigma_n^2)$. Then, $\mathbf{h}_k$ can be estimated through LS as

$$\hat{\mathbf{h}}_k = \frac{1}{\sqrt{d_k}L\sigma_p^2}\mathbf{Y}\mathbf{s}_k = \mathbf{h}_k + \frac{1}{\frac{\sqrt{P_k^{\rm ut}}}{\sigma_n^2}}\mathbf{n}_k \tag{15}$$

where $\mathbf{n}_k \sim \mathcal{CN}(\mathbf{0}, \mathbf{I})$ denotes the normalized Gaussian white noise vector.

Repeating the similar operations in (15) for all G groups yields the channel estimates for all K users. The next step is to obtain the optimal rotation ϕ_k and extract the spatial signature $\mathcal{B}_k^{\rm ro}$ of size τ that contains the maximum power of $\mathbf{F}\mathbf{\Phi}(\phi_k)\hat{\mathbf{h}}_k$, as described in (13). Then, the channel vector can be approximated by

$$\mathbf{h}_k = \mathbf{\Phi}(\phi_k)^H \mathbf{F}^H \tilde{\mathbf{h}}_k^{\rm ro} \approx \mathbf{\Phi}(\phi_k)^H [\mathbf{F}^H]_{:,\mathcal{B}_k^{\rm ro}} \left[\tilde{\mathbf{h}}_k^{\rm ro}\right]_{\mathcal{B}_k^{\rm ro},:}. \tag{16}$$

Remark 3: In contrast to existing studies [18], [20], here, we do not need channel covariance information but only spend a preamble of length GL to obtain the spatial information for all K users. As this preamble is not frequently adopted, its overhead is relatively small.

B. UL Training With User Grouping

Channel information obtained from preamble may only last for a short period, for example, one coherent time, and it should be tracked or estimated again in the later transmission. Since the channel coherence time is in the level of millisecond while a user and its surrounding obstacles may not physically change its position in the comparable time, we may treat the DOA and the AS of a user as unchanged within several or even tens of the channel coherence times. For example, if $T = 5$ ms, the maximum displacement for a user moving at 80 km/h in ten coherence times will be only 1.11 m, which means that the resultant angle variation seen by the BS is negligible. Hence, the spatial signatures $\mathcal{B}_k^{\rm ro}$ of each user could be deemed as unchanged, and only the accompanied $[\tilde{\mathbf{h}}_k^{\rm ro}]_{\mathcal{B}_k^{\rm ro},:}$ should be reestimated in the later transmission.

To remove the pilot contamination and to realize the simultaneous UL training of all K users with only τ orthogonal training sequences, we divide the users into different groups according to their spatial signatures. Specifically, the users are allocated to the same group if their spatial signatures do not overlap and are separated by a certain guard interval Ω, i.e.,

$$\mathcal{B}_k^{\rm ro} \cap \mathcal{B}_l^{\rm ro} = \emptyset, \qquad \mathrm{dist}\,(\mathcal{B}_k^{\rm ro}, \mathcal{B}_l^{\rm ro}) \geq \Omega \tag{17}$$

where $\mathrm{dist}(\mathcal{B}_1, \mathcal{B}_2) \triangleq \min |b_1 - b_2|, \forall\, b_1 \in \mathcal{B}_1, \forall\, b_2 \in \mathcal{B}_2$, and the value of Ω depends on the tolerance of users for the interference due to pilot reusing.

Remark 4: In comparison, the user grouping strategy for UL channel estimation in [18] is based on the channel covariance matrices, whereas the one in [21] is based on the sparsity pattern.

Assume that all users are divided into $G^{\rm ut}$ groups based on (17) and denote the user index set of the gth group as $\mathcal{U}_g^{\rm ut}$. We expect that $G^{\rm ut} \ll K$ since $\tau = |\mathcal{B}_k^{\rm ro}|$ is much smaller than M and users are randomly distributed in the service region. As channels of different users in the same group could be discriminated by their spatial signatures, we could assign the same training sequence to all users in one group. Without loss

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of generality, we consider $G^{\mathrm{ut}} \leq \tau$ and allocate $\mathbf{s}_i$ to the ith group.[5] Suppose all K users in G^{ut} groups send their training sequences simultaneously. Then, the received signals at the BS can be expressed as

$$\mathbf{Y} = \sum_{i=1}^{G^{\mathrm{ut}}} \sum_{k \in \mathcal{U}_i^{\mathrm{ut}}} \sqrt{d_i}\mathbf{h}_k \mathbf{s}_i^H + \mathbf{N}. \tag{18}$$

Since $\mathbf{s}_i$ are orthogonal to each other, we can extract the signals for group g as

$$\begin{aligned}\mathbf{y}_g &= \frac{1}{L\sigma_p^2}\mathbf{Y}\mathbf{s}_g = \frac{1}{L\sigma_p^2}\left(\sum_{i=1}^{G^{\mathrm{ut}}} \sum_{k \in \mathcal{U}_i^{\mathrm{ut}}} \sqrt{d_k}\mathbf{h}_k \mathbf{s}_i^H + \mathbf{N}\right)\mathbf{s}_g \\ &= \sum_{l \in \mathcal{U}_g^{\mathrm{ut}}} \sqrt{d_l}\mathbf{h}_l + \frac{1}{L\sigma_p^2}\mathbf{N}\mathbf{s}_g = \sum_{l \in \mathcal{U}_g^{\mathrm{ut}}} \sqrt{d_l}\mathbf{h}_l + \frac{1}{\frac{L\sigma_p^2}{\sigma_n^2}}\bar{\mathbf{n}}_g\end{aligned} \tag{19}$$

where $\bar{\mathbf{n}}_g \sim \mathcal{CN}(\mathbf{0}, \mathbf{I})$ is the normalized Gaussian white noise vector. Clearly, $\mathbf{y}_g$ only contains the channel vectors of users in group g, and the interference from other groups are completely eliminated by the orthogonal training. The term $\sum_{l \in \mathcal{U}_g^{\mathrm{ut}}} \sqrt{d_l}\mathbf{h}_l$ in (19) results from the reuse of the same training sequence by different users and is conventionally called *pilot contamination* [1].[6] Nevertheless, since different channel vectors in group g have different spatial signatures $\mathcal{B}_k^{\mathrm{ro}}$, we could still extract the individual channel information of each user. For example, for user k in group g, let us formulate

$$\begin{aligned}\tilde{\mathbf{y}}_{g,k}^{\mathrm{ro}} &= \frac{1}{\sqrt{d_k}}\mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{y}_g = \mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{h}_k + \sum_{l \in \left\{\mathcal{U}_g^{\mathrm{ut}} \setminus k\right\}} \sqrt{\frac{d_l}{d_k}}\mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{h}_l \\ &\quad + \frac{1}{\sqrt{\frac{P_k^{\mathrm{ut}}}{\sigma_n^2}}}\mathbf{F}\mathbf{\Phi}(\phi_k)\bar{\mathbf{n}}_g \\ &= \tilde{\mathbf{h}}_k^{\mathrm{ro}} + \sum_{l \in \left\{\mathcal{U}_g^{\mathrm{ut}} \setminus k\right\}} \sqrt{\frac{d_l}{d_k}}\mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{h}_l + \frac{1}{\sqrt{\frac{P_k^{\mathrm{ut}}}{\sigma_n^2}}}\mathbf{F}\mathbf{\Phi}(\phi_k)\bar{\mathbf{n}}_g.\end{aligned} \tag{20}$$

From SBEM, we know that $\tilde{\mathbf{h}}_k^{\mathrm{ro}}$ can be approximated by $[\tilde{\mathbf{h}}_k^{\mathrm{ro}}]_{\mathcal{B}_k^{\mathrm{ro}},:}$, and the estimate of $[\tilde{\mathbf{h}}_k^{\mathrm{ro}}]_{\mathcal{B}_k^{\mathrm{ro}},:}$ can be extracted from (20) as

$$\begin{aligned}\widehat{\left[\tilde{\mathbf{h}}_k^{\mathrm{ro}}\right]}_{\mathcal{B}_k^{\mathrm{ro}},:} &= \left[\tilde{\mathbf{y}}_{g,k}^{\mathrm{ro}}\right]_{\mathcal{B}_k^{\mathrm{ro}},:} = \frac{1}{\sqrt{d_k}}[\mathbf{I}_M]_{\mathcal{B}_k^{\mathrm{ro}},:}\mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{y}_g \\ &= \left[\tilde{\mathbf{h}}_k^{\mathrm{ro}}\right]_{\mathcal{B}_k^{\mathrm{ro}},:} + \sum_{l \in \left\{\mathcal{U}_g^{\mathrm{ut}} \setminus k\right\}} \sqrt{\frac{d_l}{d_k}}\,[\mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{h}_l]_{\mathcal{B}_k^{\mathrm{ro}},:} \\ &\quad + \frac{1}{\sqrt{\frac{P_k^{\mathrm{ut}}}{\sigma_n^2}}}\,[\mathbf{F}\mathbf{\Phi}(\phi_k)\bar{\mathbf{n}}_g]_{\mathcal{B}_k^{\mathrm{ro}},:}.\end{aligned} \tag{21}$$

[5]For $G^{\mathrm{ut}} > \tau$, similar manipulations can be repeated for the $G^{\mathrm{ut}} - \tau$ groups.

[6]This interuser interference stems from the pilot reuse in the same cell and thus can be specified as the intracell pilot contamination [32]. On the other hand, for the common intercell pilot contamination, our method can be similarly applied.

Since $\mathcal{B}_l^{\mathrm{ro}}$ and $\mathcal{B}_k^{\mathrm{ro}}$ are separated at least by one guard interval Ω, we know that the entries of $[\mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{h}_l]_{\mathcal{B}_k^{\mathrm{ro}},:}$ in (21) are very small. Then, the estimate of $\tilde{\mathbf{h}}_k^{\mathrm{ro}}$ could be approximated as

$$\hat{\tilde{\mathbf{h}}}_k^{\mathrm{ro}} = \left[\mathbf{0}^T \quad \widehat{\left[\tilde{\mathbf{h}}_k^{\mathrm{ro}}\right]}_{\mathcal{B}_k^{\mathrm{ro}},:}^H \quad \mathbf{0}^T\right]^H \tag{22}$$

where the two zero vectors $\mathbf{0}$ have appropriate sizes. Hence, the channel estimate of user k could be computed as

$$\hat{\mathbf{h}}_k = \mathbf{\Phi}(\phi_k)^H\mathbf{F}^H\hat{\tilde{\mathbf{h}}}_k^{\mathrm{ro}} = \mathbf{\Phi}(\phi_k)^H[\mathbf{F}^H]_{:,\mathcal{B}_k^{\mathrm{ro}}}[\mathbf{I}_M]_{\mathcal{B}_k^{\mathrm{ro}},:}\mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{y}_g. \tag{23}$$

The mean square error (MSE) of $\hat{\mathbf{h}}_k$ in (23) is expressed as

$$\begin{aligned}\mathrm{MSE}_k^u &= \mathbb{E}\left\{\|\mathbf{h}_k - \hat{\mathbf{h}}_k\|^2\right\} = \mathbb{E}\left\{\left\|\mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{h}_k - \mathbf{F}\mathbf{\Phi}(\phi_k)\hat{\mathbf{h}}_k\right\|^2\right\} \\ &= \left\|\left[\tilde{\mathbf{h}}_k^{\mathrm{ro}}\right]_{\Xi \setminus \mathcal{B}_k^{\mathrm{ro}},:}\right\|^2 + \left\|\sum_{l \in \mathcal{U}_g^{\mathrm{ut}} \setminus \{k\}} [\mathbf{F}\mathbf{\Phi}(\phi_k)\mathbf{h}_l]_{\mathcal{B}_k^{\mathrm{ro}},:}\right\|^2 + \frac{\tau}{\frac{P_k^{\mathrm{ut}}}{\sigma_n^2}}\end{aligned} \tag{24}$$

where Ξ denotes the complete index set $\{0, \ldots, M-1\}$.

It is easy to infer from (24) that the channel estimation error is composed of three parts. The first part is obviously the truncation error from SBEM that only keeps τ DFT points in $\tilde{\mathbf{h}}_k^{\mathrm{ro}}$. The channel leakage from other users in the same group to the current $\mathcal{B}_k^{\mathrm{ro}}$ brings the second error term, which formulates *remaining* pilot contamination. The noise term accounts for the last error part, which is proportional to τ [33]. The truncation error will be reduced if the AS of user k becomes smaller or if the value of τ increases. However, AS is the inherent property of the environment, and τ is generally regulated by the standards, both being uncontrollable. Nevertheless, the interuser interferences can be reduced when Ω is set to be larger. Hence, if G^{ut} is smaller than τ, then we should reformulate exactly $G^{\mathrm{ut}} = \tau$ groups with evenly distributed users in each group such that the second error term can be reduced.

C. Angle Reciprocity and DL Channel Representation

Denote the DL channel from the BS to user k as $\mathbf{g}_k^H \in \mathbb{C}^{1\times M}$. Similar to (1), $\mathbf{g}_k \in \mathbb{C}^{M\times 1}$ can be modeled as

$$\mathbf{g}_k = \frac{1}{\sqrt{P}}\sum_{p=1}^{P}\beta_{kp}\mathbf{a}(\vartheta_{kp}) \tag{25}$$

where $\mathbf{a}(\vartheta_{kp})$ is the steering vector defined in (2), but with different DL carrier wavelength λ_2 if FDD mode is selected; ϑ_{kp} is the DOD of the pth ray that could arrive at user k; and β_{kp} is the corresponding complex gain. All other parameters have the same definitions as in (1).

Since the propagation path of electromagnetic wave is reciprocal, we know that only the signal wave that physically reverses the UL path can reach the user during the DL period. Hence, DL signals that could effectively arrive at the user should have the same DOD spread as the UL DOA spread, that is, the angle range of ϑ_{kp} is the same as the angle range of

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θ_{kp}. We call this property of the wireless channel as *angle reciprocity*.[7] For TDD systems, the channel gain is also reciprocal such that the overall channel is reciprocal.

Hence, the DL channel $\mathbf{g}_k$ can be also approximated by SBEM with appropriate spatial signatures $\overline{\mathcal{B}_k^{\mathrm{ro}}}$ and rotation parameter $\bar{\phi}_k$, i.e.,

$$\begin{aligned}\mathbf{g}_k &= \mathbf{\Phi}(\bar{\phi}_k)^H \mathbf{F}^H \tilde{\mathbf{g}}_k^{\mathrm{ro}} \approx \mathbf{\Phi}(\bar{\phi}_k)^H [\mathbf{F}^H]_{:,\overline{\mathcal{B}_k^{\mathrm{ro}}}} [\tilde{\mathbf{g}}_k^{\mathrm{ro}}]_{\overline{\mathcal{B}_k^{\mathrm{ro}}},:} \\ &= \sum_{q \in \overline{\mathcal{B}_k^{\mathrm{ro}}}} \tilde{g}_{k,q}^{\mathrm{ro}} \mathbf{\Phi}(\bar{\phi}_k)^H \mathbf{f}_q \end{aligned} \tag{26}$$

where $\tilde{g}_{k,q}^{\mathrm{ro}} \triangleq [\tilde{\mathbf{g}}_k^{\mathrm{ro}}]_q$ denotes the qth element of $\tilde{\mathbf{g}}_k^{\mathrm{ro}} = \mathbf{F}\mathbf{\Phi}(\bar{\phi}_k)\mathbf{g}_k$.

Based on the angle reciprocity, $\overline{\mathcal{B}_k^{\mathrm{ro}}}$ can be determined by $\mathcal{B}_k^{\mathrm{ro}}$. Specifically, according to Lemma 2, we will have

$$\sin\theta_{kp} = \frac{q\lambda_1}{Md} = \frac{q'\lambda_2}{Md}, \text{ with } q \in \mathcal{B}_k^{\mathrm{ro}},\ q' \in \overline{\mathcal{B}_k^{\mathrm{ro}}} \tag{27}$$

where λ_1 and λ_2 denote the UL/DL carrier wavelengths, respectively. Then, the integer set $\overline{\mathcal{B}_k^{\mathrm{ro}}}$ can be expressed as $\overline{\mathcal{B}_k^{\mathrm{ro}}} = \{q'_{\min}, q'_{\min}+1, \ldots, q'_{\max}\}$, with

$$q'_{\min} = \left\lfloor \frac{\lambda_1}{\lambda_2} q_{\min} \right\rfloor \qquad q'_{\max} = \left\lceil \frac{\lambda_1}{\lambda_2} q_{\max} \right\rceil \tag{28}$$

where $q_{\min} \le q \le q_{\max}\ \forall\, q \in \mathcal{B}_k^{\mathrm{ro}}$. Similarly, $\bar{\phi}_k$ can be determined as $\bar{\phi}_k = (\lambda_1/\lambda_2)\phi_k$.

D. DL Training With User Grouping

From SBEM (26), DL channel estimation for each user only needs to estimate τ parameters. To reuse the overall τ orthogonal training sequences, let us divide K users into different groups. Since the DL training for users with identical spatial signatures can be carried out with the same beamforming vectors simultaneously, we propose to first gather users with identical spatial signatures $\overline{\mathcal{B}_k^{\mathrm{ro}}}$ into the same cluster. Then, these clusters are divided into different groups such that the spatial signatures of the clusters in the same group do not overlap and are separated by a certain guard interval, as done in (17).

Remark 5: In contrast, the user grouping strategy for DL channel estimation in [20] is based on the channel covariance matrices, whereas the one in [17] is based on the sparsity pattern.

For ease of exposition, assume that all K users are divided into G^{dt} groups, and denote the user index set of the gth group as $\mathcal{U}_g^{\mathrm{dt}}$. Let us then take the training of the group g, for example, to describe the DL channel estimation algorithm. Based on SBEM, the effective transmission between τ DFT points in $\overline{\mathcal{B}_k^{\mathrm{ro}}}$ and each user formulates a virtual $\tau \times 1$ multiple-input–single-output (MISO) DL system. Hence, to estimate the τ coefficients $[\tilde{\mathbf{g}}_k^{\mathrm{ro}}]_{\overline{\mathcal{B}_k^{\mathrm{ro}}}}$ for each user, we need to transmit τ orthogonal training sequences from the corresponding τ beams $\mathbf{f}_q,\ q \in \overline{\mathcal{B}_k^{\mathrm{ro}}}$. We then select the orthogonal training matrix for user k as $\mathbf{S}_k = \varpi_k \mathbf{S}^H \in \mathbb{C}^{\tau \times L}$, where the scalar variable ϖ_k is used to satisfy the transmit power constraint $\mathrm{tr}\{\mathbf{S}_k\mathbf{S}_k^H\} \le P_k^{\mathrm{dt}}$, and thus, $\varpi_k = \sqrt{P_k^{\mathrm{dt}}/\tau L\sigma_p^2}$. Note that the same orthogonal training matrix $\mathbf{S}^H \in \mathbb{C}^{\tau \times L}$ is reused by different users over their own spatial signatures in the same group.

Then, the received signals at user k of group g can be expressed as

$$\begin{aligned}\mathbf{y}_k^H &= \mathbf{g}_k^H \left(\sum_{l \in \mathcal{U}_g^{\mathrm{dt}}} \mathbf{\Phi}(\bar{\phi}_l)^H [\mathbf{F}^H]_{:,\overline{\mathcal{B}_l^{\mathrm{ro}}}} \mathbf{S}_l \right) + \mathbf{n}_k^H \\ &= \sum_{l \in \mathcal{U}_g^{\mathrm{dt}}} \left([\mathbf{F}]_{\overline{\mathcal{B}_l^{\mathrm{ro}}},:} \mathbf{\Phi}(\bar{\phi}_l) \mathbf{g}_k \right)^H \mathbf{S}_l + \mathbf{n}_k^H \\ &= [\tilde{\mathbf{g}}_k^{\mathrm{ro}}]_{\overline{\mathcal{B}_k^{\mathrm{ro}}},:}^H \mathbf{S}_k + \sum_{l \in \{\mathcal{U}_g^{\mathrm{dt}} \setminus k\}} \left[\mathbf{F}\mathbf{\Phi}(\bar{\phi}_l)\mathbf{g}_k \right]_{\overline{\mathcal{B}_l^{\mathrm{ro}}},:}^H \mathbf{S}_l + \mathbf{n}_k^H \end{aligned} \tag{29}$$

where $\mathbf{n}_k^H \in \mathbb{C}^{1\times L}$ is the noise vector, with its elements distributed as $\mathcal{CN}(0, \sigma_n^2)$. The DL channel of user k can be estimated through the LS method as

$$\begin{aligned}\widehat{[\tilde{\mathbf{g}}_k^{\mathrm{ro}}]}_{\overline{\mathcal{B}_k^{\mathrm{ro}}},:}^H &= \mathbf{y}_k^H \mathbf{S}_k^\dagger = [\tilde{\mathbf{g}}_k^{\mathrm{ro}}]_{\overline{\mathcal{B}_k^{\mathrm{ro}}},:}^H \\ &\quad + \sum_{l \in \{\mathcal{U}_g^{\mathrm{dt}} \setminus k\}} \frac{\varpi_l}{\varpi_k} \left[\mathbf{F}\mathbf{\Phi}(\bar{\phi}_l)\mathbf{g}_k \right]_{\overline{\mathcal{B}_l^{\mathrm{ro}}},:}^H + \mathbf{n}_k^H \mathbf{S}_k^\dagger. \end{aligned} \tag{30}$$

Since $\overline{\mathcal{B}_l^{\mathrm{ro}}}$ is separated at least one guard interval from $\overline{\mathcal{B}_k^{\mathrm{ro}}}$, $[\mathbf{F}\mathbf{\Phi}(\bar{\phi}_l)\mathbf{g}_k]_{\overline{\mathcal{B}_l^{\mathrm{ro}}},:}$ will be very small compared to $[\tilde{\mathbf{g}}_k^{\mathrm{ro}}]_{\overline{\mathcal{B}_k^{\mathrm{ro}}},:}^H$. Then, the estimate of $\tilde{\mathbf{g}}_k^{\mathrm{ro}}$ can be expressed as $\hat{\tilde{\mathbf{g}}}_k^{\mathrm{ro}} = [\mathbf{0}^T\ \widehat{[\tilde{\mathbf{g}}_k^{\mathrm{ro}}]}_{\overline{\mathcal{B}_k^{\mathrm{ro}}},:}^H\ \mathbf{0}^T]^H$,, where the two zero vectors $\mathbf{0}$ have appropriate sizes. Following (26), the channel estimate for user k in group g is obtained as

$$\hat{\mathbf{g}}_k = \mathbf{\Phi}(\bar{\phi}_k)^H \mathbf{F}^H \hat{\tilde{\mathbf{g}}}_k^{\mathrm{ro}} = \mathbf{\Phi}(\bar{\phi}_k)^H [\mathbf{F}^H]_{:,\overline{\mathcal{B}_k^{\mathrm{ro}}}} \left(\mathbf{S}_k^H\right)^\dagger \mathbf{y}_k. \tag{31}$$

After channel estimation, each user feeds back τ components $[\hat{\tilde{\mathbf{g}}}_k^{\mathrm{ro}}]_{\overline{\mathcal{B}_k^{\mathrm{ro}}},:}$ to the BS such that the BS can perform the optimal user scheduling and power allocation for the subsequent DL data transmission. Note that these τ components may be further quantified for feedback in different ways, such as in [37] and [38].

Similar to the UL analysis, the DL MSE of the LS estimator for user k can be expressed as

$$\begin{aligned}\mathrm{MSE}_k^d &= \mathbb{E}\left\{ \|\mathbf{g}_k - \hat{\mathbf{g}}_k\|^2 \right\} = \mathbb{E}\left\{ \left\| \tilde{\mathbf{g}}_k^{\mathrm{ro}} - \hat{\tilde{\mathbf{g}}}_k^{\mathrm{ro}} \right\|^2 \right\} \\ &= \left\| \tilde{\mathbf{g}}_k^{\mathrm{ro}} - \left[\mathbf{0}^H \ \left[\hat{\tilde{\mathbf{g}}}_k^{\mathrm{ro}}\right]_{\overline{\mathcal{B}_k^{\mathrm{ro}}},:}^H \ \mathbf{0}^H \right]^H \right\|^2 = \left\| [\tilde{\mathbf{g}}_k^{\mathrm{ro}}]_{\Xi \setminus \overline{\mathcal{B}_k^{\mathrm{ro}}},:} \right\|^2 \\ &\quad + \left\| \sum_{l \in \{\mathcal{U}_g^{\mathrm{dt}} \setminus k\}} \frac{\varpi_l}{\varpi_k} \left[\mathbf{F}\mathbf{\Phi}(\bar{\phi}_l)\mathbf{g}_k \right]_{\overline{\mathcal{B}_l^{\mathrm{ro}}},:}^H \right\|^2 + \frac{\tau^2}{\frac{P_k^{\mathrm{dt}}}{\sigma_n^2}} \end{aligned} \tag{32}$$

[7] If the frequency of the DL channel is not too far from that of the UL channel, e.g., less than several gigahertz, then the reciprocity between DOD and DOA accurately holds. The reason is that, for the typical transmission environment, the relative permittivity and the conductivity of the obstacles do not change in the scale of several dozens of gigahertz [34]. Hence, the reflection and deflection properties of signals with less than several gigahertz frequency differences are almost identical in such propagation environments [35], [36].

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which is also composed of three parts. The first part is, again, the truncation error from SBEM with only τ spatial signatures. The channel leakage of SBEM from user k itself results in the second term, which is different from the UL case. The last error term comes from the noise and is proportional to τ^2, which is consistent with the result in the conventional $\tau \times 1$ MISO DL system [33]. The self-interference due to pilot reuse will be reduced when the number of users in the same group decreases and the guard interval becomes larger.

E. Blind DOA Tracking During Data Transmission

The DOA information and the spatial signatures of users can be blindly tracked even during the data transmission. Specifically, after UL user grouping and training, different groups of users will be scheduled into separate time slots for UL data transmission. Then, the received data $\mathbf{y}_g^{\mathrm{ul}} \in \mathbb{C}^{M\times 1}$ at the BS for group g can be expressed as

$$\mathbf{y}_g^{\mathrm{ul}} = \sum_{k\in\mathcal{U}_g^{\mathrm{ut}}} \mathbf{h}_k x_k + \mathbf{n}_g^{\mathrm{ul}} \tag{33}$$

where the scale variable x_k is the UL data symbol from user k, and $\mathbf{n}_g^{\mathrm{ul}} \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2\mathbf{I})$ is the noise vector. Since users in the same group have disjoint initial spatial signatures, the DFT of $\mathbf{y}_g^{\mathrm{ul}}$ will have several nonoverlapped segments of significant components, which directly provides us the new spatial signatures of each user. Based on this, the users' spatial signatures can be blindly tracked until the distance between two signatures becomes less than the guard interval. Obviously, this data-aided blind tracking scheme reduces the frequency of sending the preamble.

IV. Data Transmission With User Scheduling

After obtaining the spatial signatures and channel gains of all users, we may schedule the users into different groups to enhance the spectral efficiency during the subsequent data transmission period. To achieve this, users in the same group should have nonoverlapping spatial signatures such that the interuser interference could be reduced. Meanwhile, we try to maximize the achievable rate for each group under given power constraint. In the following discussion, we will focus on the DL case due to space limitations, whereas the UL case can be analyzed in a similar manner.

Assume that users are scheduled into G^{dd} groups for data transmission and the user index set of the gth group is denoted by $\mathcal{U}_g^{\mathrm{dd}}$, $g = 1, 2, \ldots, G^{\mathrm{dd}}$. For user k in the gth group, its DL received signal can be expressed as

$$\begin{aligned} y_k^{\mathrm{dl}} &= \sqrt{\rho_k}\mathbf{g}_k^H \mathbf{w}_k x_k + \sum_{l\in\{\mathcal{K}_g\setminus k\}} \sqrt{\rho_l}\mathbf{g}_k^H \mathbf{w}_l x_l + n_k \\ &= \sqrt{\rho_k}\left(\hat{\mathbf{g}}_k^H + \Delta\mathbf{g}_k^H\right)\mathbf{w}_k x_k \\ &\quad + \sum_{l\in\{\mathcal{K}_g\setminus k\}} \sqrt{\rho_l}\left(\hat{\mathbf{g}}_k^H + \Delta\mathbf{g}_k^H\right)\mathbf{w}_l x_l + n_k \end{aligned} \tag{34}$$

where channel estimation error $\Delta\mathbf{g}_k$ is defined as $\Delta\mathbf{g}_k \triangleq \mathbf{g}_k - \hat{\mathbf{g}}_k$; x_k and $\mathbf{w}_k$ denote the data symbol and the corresponding DL beamforming vector, respectively; $\rho_k > 0$ denotes the transmit power; and $n_k \sim \mathcal{CN}(0,1)$ is the received noise.

For DL massive MIMO systems, the linear matched filter beamforming was already shown to be a good candidate [1], that is

$$\mathbf{w}_k = \frac{\hat{\mathbf{g}}_k}{\|\hat{\mathbf{g}}_k\|} = \frac{\sum_{q\in\mathcal{B}_k^{\mathrm{ro}}} \hat{\tilde{g}}_{k,q}^{\mathrm{ro}} \mathbf{\Phi}(\phi_k)^H \mathbf{f}_q}{\sum_{q\in\mathcal{B}_k^{\mathrm{ro}}} \left|\hat{\tilde{g}}_{k,q}^{\mathrm{ro}}\right|}. \tag{35}$$

Interestingly, the structure of $\mathbf{w}_k$ indicates that the overall beamforming is composed of τ orthogonal subbeams that correspond to τ spatial signatures of user k, and the beam gains are exactly selected as their estimated channel gains $\hat{\tilde{g}}_{k,q}^{\mathrm{ro}}$. Since users in the same group have nonoverlapped spatial signatures, i.e., $\mathcal{B}_k^{\mathrm{ro}} \cap \mathcal{B}_l^{\mathrm{ro}} = \emptyset$, it directly leads to $\mathbf{w}_k^H \mathbf{w}_l = \hat{\mathbf{g}}_k^H \hat{\mathbf{g}}_l = 0$. Note that the beam orthogonality in our proposed strategy always holds for any finite value of M, as opposed to many other studies, which require $M \to \infty$ [18], [39], [40].

Then, (34) can be simplified as

$$y_k^{\mathrm{dl}} = \sqrt{\rho_k}\|\hat{\mathbf{g}}_k\| x_k + \Delta\mathbf{g}_k^H \sum_{l\in\mathcal{K}_g} \sqrt{\rho_l} \frac{\hat{\mathbf{g}}_l}{\|\hat{\mathbf{g}}_l\|} x_l + n_k. \tag{36}$$

Since we do not assume channel statistics, we cannot apply the conventional way [41] to characterize the lower bound of achievable throughput. Nevertheless, in practical case, the estimated channel $\hat{\mathbf{g}}_k$ will be used as if it is the true value of $\mathbf{g}_k$. Hence, we will treat the second term in (36) as Gaussian noise with the same variance, which is a worst case, and then solve the following optimization problem to approximately maximize the throughput of each group:

$$\begin{aligned} \max_{\{\rho_k\}} \quad & R\left(\mathcal{U}_g^{\mathrm{dd}}|\mathcal{P}\right) \triangleq \sum_{k\in\mathcal{U}_g^{\mathrm{dd}}} \log_2(1+\gamma_k) \\ \text{s.t.} \quad & \sum_{k\in\mathcal{U}_g^{\mathrm{dd}}} \rho_k \le \mathcal{P} \end{aligned} \tag{37}$$

where the equivalent signal-to-noise ratio (SNR) γ_k is given as

$$\gamma_k = \frac{\rho_k \left\| \left[\hat{\mathbf{g}}_k^{\mathrm{ro}}\right]_{\mathcal{B}_k^{\mathrm{ro}},:} \right\|^2}{1 + \sum_{l\in\mathcal{K}_g} \rho_l \left|\Delta\mathbf{g}_k^H \hat{\mathbf{g}}_l\right|^2 \Big/ \|\hat{\mathbf{g}}_l\|^2} \tag{38}$$

and $\mathcal{P}$ is the total power constraint normalized by white noise power for this group. Solutions to (37) can be obtained by the standard water-filling algorithm [42].

We then provide a greedy user scheduling algorithm, where the user with the strongest channel gain will be first scheduled, and then the other users with nonoverlapping spatial signatures can join the same group only if the achievable sum rate of the whole group increases afterward. The detailed steps can be found in Algorithm 1, where we assume that the average power constraint for each single user is ρ, and thus, the power constraint $\mathcal{P}$ for each group is adjusted dynamically to be proportional to the final number of users in each group.

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Algorithm 1 User Scheduling for Data Transmission

- **Initialize**: Input $\hat{\mathbf{g}}_k$, $\mathcal{B}_k^{\mathrm{ro}}$ and the average transmit power constraint for a single user ρ.
- **Step 1**: Calculate the Euclidean norm of the estimated channel vectors, i.e., $\|\hat{\mathbf{g}}_l\| = \|[\hat{\mathbf{g}}_l^{\mathrm{ro}}]_{\mathcal{B}_l^{\mathrm{ro}},:}\|$, for all users.
- **Step 2**: Initialize $g = 1$, $\mathcal{P} = 0$, $\mathcal{U}_g^{\mathrm{dd}} = \emptyset$, $R(\mathcal{U}_g^{\mathrm{dd}}|\mathcal{P}) = 0$, and the remaining user set $\mathcal{U}_r = \{1, \ldots, K\}$.
- **Step 3**: For the gth group, select the user in $\mathcal{U}_r$ with the maximal norm of channel, $l' = \arg\max_{l \in \mathcal{U}_r} \|\hat{\mathbf{g}}_l\|$. Set $\mathcal{P} = \rho$, $\mathcal{U}_g^{\mathrm{dd}} = \mathcal{U}_g^{\mathrm{dd}} \cup \{l'\}$ and $\mathcal{U}_r = \mathcal{U}_r \setminus \{l'\}$. Calculate $R(\mathcal{U}_g^{\mathrm{dd}}|\mathcal{P})$ according to (37).
- **Step 4**: Select all those users in $\mathcal{U}_r$ whose spatial signatures are nonoverlapping with users in $\mathcal{U}_g^{\mathrm{dd}}$, and denote them by $\mathcal{U}_g'$, i.e.,

$$\mathcal{U}_g' = \left\{ m \in \mathcal{U}_r | \mathcal{B}_m^{\mathrm{ro}} \cap \mathcal{B}_l^{\mathrm{ro}} = \emptyset \text{ and } \mathrm{dist}\left(\mathcal{B}_m^{\mathrm{ro}}, \mathcal{B}_l^{\mathrm{ro}}\right) \geq \Omega, \forall\, l \in \mathcal{U}_g^{\mathrm{dd}} \right\}.$$

- **Step 5**: If $\mathcal{U}_g' \neq \emptyset$, set $\mathcal{P}' = \mathcal{P} + \rho$, and find a user m' in $\mathcal{U}_g'$, such that

$$m' = \arg\max_{m \in \mathcal{U}_g'} \; R\left(\mathcal{U}_g^{\mathrm{dd}} \cup \{m\} \middle| \mathcal{P}'\right).$$

  If $R(\mathcal{U}_g^{\mathrm{dd}} \cup \{m\}|\mathcal{P}') \geq R(\mathcal{U}_g^{\mathrm{dd}}|\mathcal{P})$, set $\mathcal{U}_g^{\mathrm{dd}} = \mathcal{U}_g^{\mathrm{dd}} \cup \{m'\}$, $\mathcal{P} = \mathcal{P}'$, $\mathcal{U}_r = \mathcal{U}_r \setminus \{m'\}$ and go to **Step 4**; Else, go to **Step 6**.
- **Step 6**: Store $\mathcal{U}_g^{\mathrm{dd}}$ and $R(\mathcal{U}_g^{\mathrm{dd}}|\mathcal{P})$. If $\mathcal{U}_r \neq \emptyset$, let $g = g + 1$, go to **Step 3**; Else, go to **Step 7**.
- **Step 7**: When the algorithm is stopped, the minimal number of user group G^{dd} is set as the current g, and the optimal user scheduling result is accordingly given by $\mathcal{U}_1, \ldots, \mathcal{U}_{G^{\mathrm{dd}}}$.

V. Simulations

Here, we demonstrate the effectiveness of the proposed strategy through numerical examples. We select $M = 128$ and $d = \lambda/2$ and consider $K = 32$ active users that are gathered into four spatially distributed clusters around the mean directions of $\{-48.59°, -14.48°, 14.48°, 48.59°\}$, respectively. The channel vectors of different users are formulated according to (1), where $P = 100$, and α_{kp} is independent and identically distributed as $\mathcal{CN}(0, 1)$ for all rays and all users; θ_{kp} is uniformly distributed inside $[\theta_k - \Delta\theta_k, \theta_k + \Delta\theta_k]$, where the AS is supposed to be $\Delta\theta_k = 2°$ for all users. The system coherence interval is set as $T = 128$, and the default value of τ is assumed to be $\tau = 16$, which is only one eighth of the antenna number, and the guard interval for user grouping is set as $\Omega = \tau/4$. The length of pilot L should satisfy $16 \leq L \leq 128$ and is then taken as $L = 16, 32$, and 64, respectively.[8] The SNR is defined as $\rho = \sigma_p^2/\sigma_n^2$. The performance metric of the channel estimation is taken as the average individual MSE, i.e.,

$$\mathrm{MSE} \triangleq \frac{1}{K} \sum_{k=1}^{K} \frac{\|\mathbf{h}_k - \hat{\mathbf{h}}_k\|^2}{\|\mathbf{h}_k\|^2}.$$

[8]As explained earlier, each cell is allocated the minimum pilot sequences required, with extra ones assigned for other cells. Thus, here, $\tau \leq L$ is required.

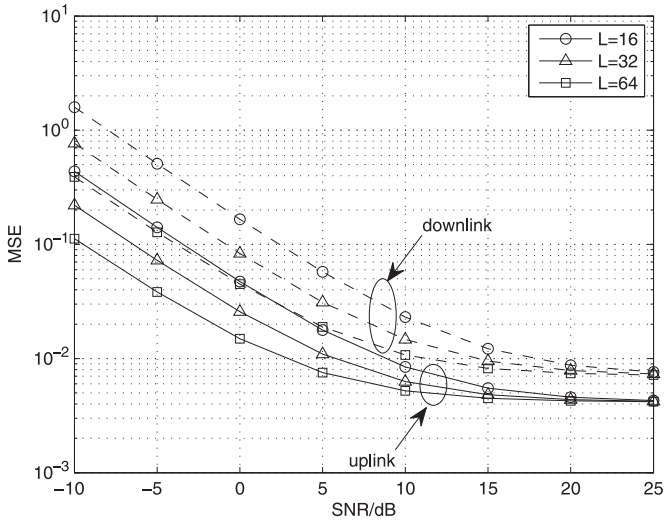


Fig. 5. Comparison of UL/DL MSE performances of the proposed SBEM method with $\tau = 16$ and $L = 16, 32$, and 64, respectively.

In all examples, the spatial information of users is estimated from the preamble. When $\tau = 16$, the 32 users are scheduled into two groups, whereas when $\tau = 8$, the 32 users are scheduled into four groups, such that the orthogonal training can be applied to obtain the spatial signatures.

Fig. 5 illustrates the MSE performances of UL/DL training in (23) and (31), respectively, as a function of SNR with different training sequence length L. The total power for both the UL and DL training is constrained to $P_k^{\mathrm{ut}} = P_k^{\mathrm{dt}} = L\rho$ for all users at a given SNR ρ. For the UL training, $K = 32$ users are divided into $G^{\mathrm{ut}} = 16$ groups. All these 16 groups can be scheduled in the same training length L with $\tau = 16$ available orthogonal training sequences. On the other hand, for the DL training, $K = 32$ users are gathered into four clusters and are assigned into $G^{\mathrm{dt}} = 1$ group, i.e., they can be scheduled simultaneously in the same training length L as well. It is shown in Fig. 5 that, when L increases, the MSE performances of UL/DL can be improved, since the total training power is proportional to L. It is also shown that, as the SNR increases, there is the same error floor for all values of L during both the UL and DL training. This phenomenon is not unexpected due to the truncation error of SBEM from the real channel and can be also observed in temporal BEM [30]. Meanwhile, since the truncation error is only related with the effective expansion number τ, the error floors will remain the same for different L. Moreover, it can be seen that the UL MSE performances are generally better than those of DL for any SNR and L, and each of the UL MSE curves is almost parallel to the corresponding DL one with a fixed gap between them. This can be inferred by comparing the noise terms of (24) and (32), where the noise power included in the UL training is only proportional to τ, whereas it is proportional to τ^2 for the DL training.

Figs. 6 and 7 compare the proposed channel estimation with the convention LS method for both the UL and DL cases. To apply the conventional method, $K = 32$ orthogonal training sequences are used for the UL case, whereas a 128×128 orthogonal training matrix is used for the DL case. To provide

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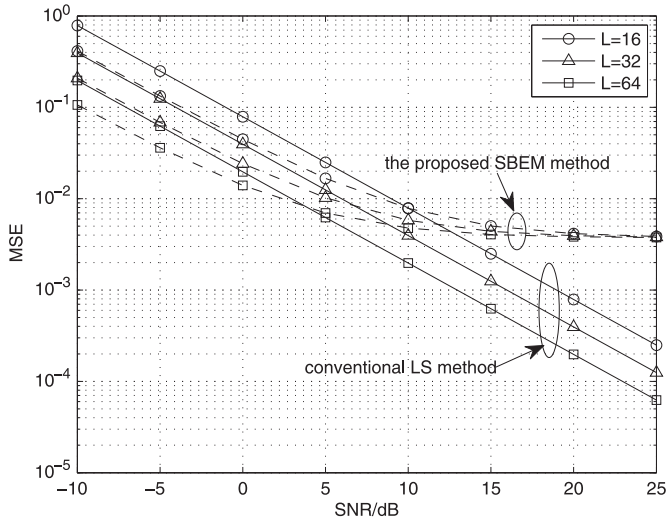


Fig. 6. UL MSE performance comparison of the proposed SBEM method and the conventional LS method, with $\tau = 16$ and $L = 16, 32$, and 64, respectively.

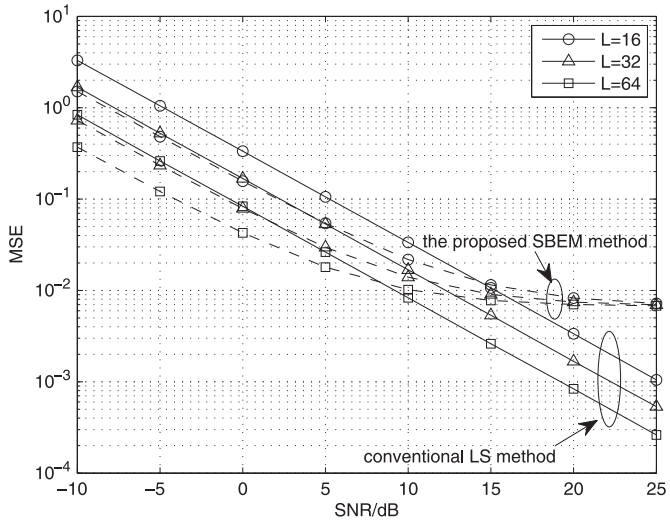


Fig. 7. DL MSE performance comparison of the proposed SBEM method and the conventional LS method, with $\tau = 16$ and $L = 16, 32$, and 64, respectively.

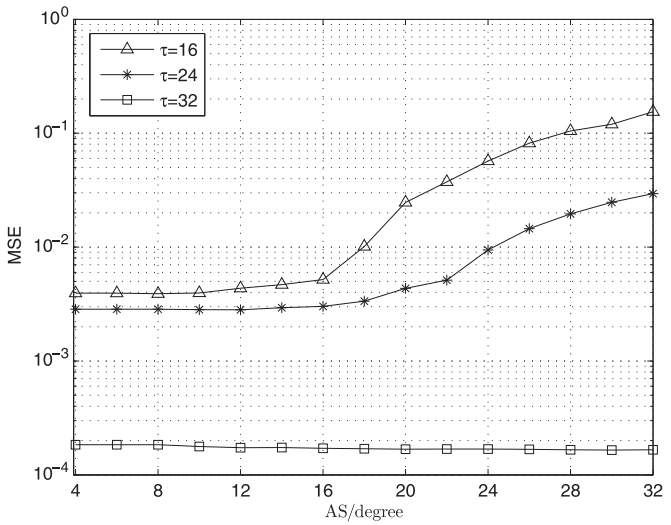


Fig. 8. UL MSE as a function of AS, with $K = 32$; $\tau = 16, 24$, and 32; $L = 64$; and SNR $= 20$ dB.

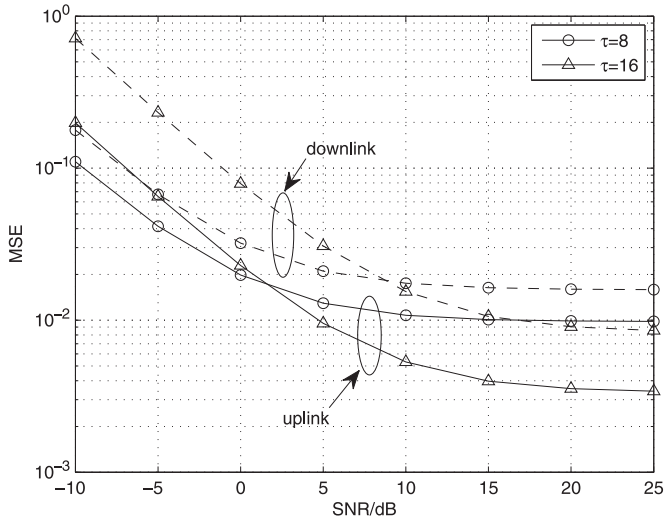


Fig. 9. Comparison of UL/DL MSE performances with $L = 32$ and $\tau = 8$ and 16 , respectively.

a fair comparison, for any given ρ and L, the UL training power is kept the same $P_k^{\mathrm{ut}} = L\rho$ for both methods, and the total DL training power for both methods are constrained as $\sum_{k=1}^{K} P_k^{\mathrm{dt}} = KL\rho$. Note that, to compare with LS, we set $T = 128$, and under such conditions, LS will consume all $T = 128$ symbols for training with no remaining time for data transmission, whereas the UL and DL training overheads of the proposed scheme are only $\lceil G^{\mathrm{ut}}/\tau \rceil L \ll T$ and $G^{\mathrm{dt}} L \ll T$, respectively. It is seen that, when the sufficient number of orthogonal training is available and when the computational complexity is acceptable, the conventional LS method does not have the error floor for both the UL and DL cases. Nevertheless, it is interesting to observe that the channel estimation from SBEM outperforms the conventional method when the SNR is relatively low. The reasons can be found from (24) and (32), where the proposed method only involves τ components of the noise vector, whereas the conventional LS method includes the whole noise power.

Fig. 8 displays the UL MSE of the proposed SBEM as a function of AS. It can be seen that the UL training by $\tau = 16$ orthogonal sequences can achieve a good performance for a relatively large AS range, for example, 16°. When the AS is increased, the number of unknowns would also increase, and we need to enhance the number of orthogonal training sequences to get satisfactory performances, like the case of $\tau = 24$. Furthermore, if the number of orthogonal training sequences is further incremented to $\tau = K = 32$, then our SBEM is equivalent to the conventional LS and the MSE performance is not affected by the AS.

Fig. 9 displays the UL/DL MSEs as a function of SNR for $\tau = 8$ and 16, respectively, with $L = 32$. The total power for both the UL and DL training is constrained to $L\rho$ for all users at a given SNR ρ. It is seen that as τ increases, the error floors of

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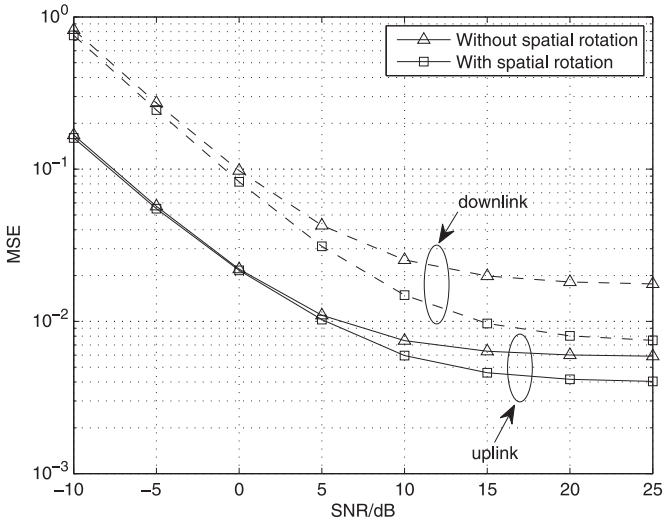


Fig. 10. Comparison of UL/DL MSE with and without spatial rotation, where $\tau = 16$, and $L = 32$.

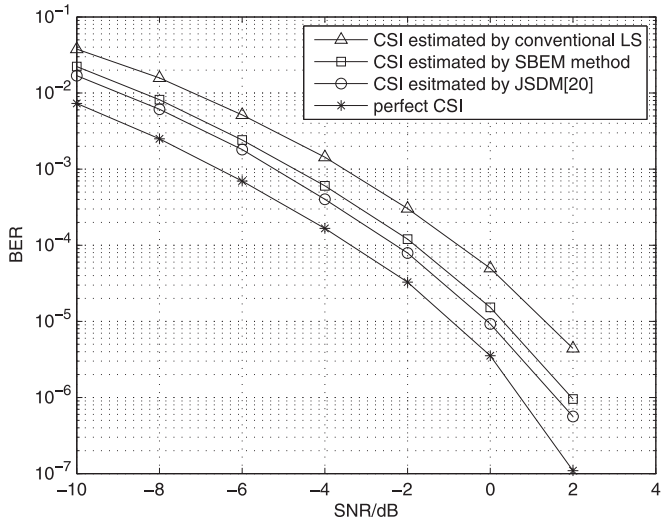


Fig. 12. Comparison of BER performances with perfect CSI, CSI estimated by JSDM [20], CSI estimated by SBEM, and CSI estimated by the conventional LS method, respectively, where $\tau = 16$, and $L = 32$.

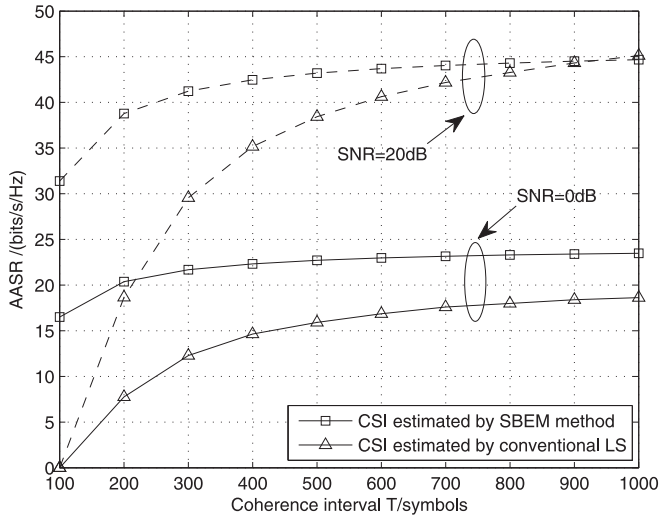


Fig. 11. AASR of the proposed SBEM and the conventional LS as a function of coherence interval T with $L = 32$.

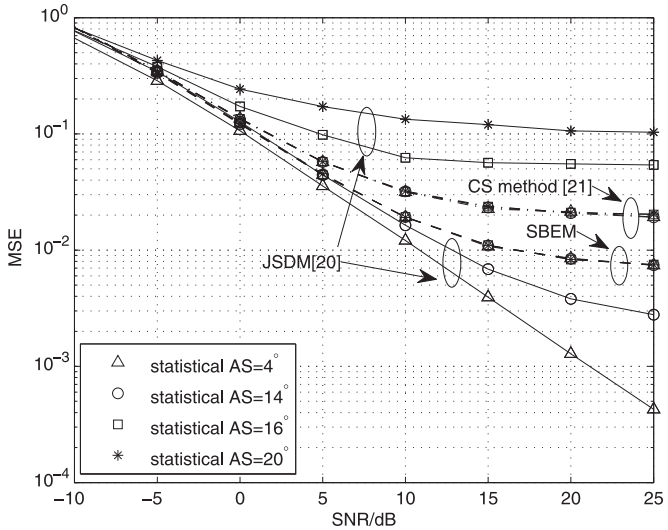


Fig. 13. DL MSE performances of SBEM, JSDM [20], and the CS-based method [21] under the case of user mobility, where $\tau = 16$, and $L = 32$.

UL and DL MSEs will decline as expected. Interestingly, for the DL training, larger τ will perform better than smaller τ at higher SNR, but will perform worse at lower SNR. The reasons might be inferred from (32), where the three error components are all closely related to the value of $\tau = |\mathcal{B}_k^{\mathrm{ro}}|$. When SNR decreases, the noise component will dominate (32), and thus, larger τ will bring more error. On the other hand, for higher SNR, the truncation error will dominate (32), and thus, smaller τ will bring more error. Similar phenomena can be also observed for the UL training, which can be explained in the same manner over (24).

As spatial rotation (13) is a key technique of our newly proposed strategy, it is then of interest to see how spatial rotation helps improve the performance. We display the channel estimation for both UL and DL with and without spatial rotation in Fig. 10. When the spatial rotation is not adopted, the shift parameter ϕ_k in (13) is set as $\phi_k = 0$, and then, the channel estimation procedures remain the same. It is clearly shown in Fig. 10 that the spatial rotation will improve the channel estimation accuracy for both UL and DL mainly at high SNRs, i.e., when error floor happens. As explained in *Property 1*, spatial rotation concentrates more channel power on fewer DFT points, reduces the power leakage outside $\mathcal{B}_k^{\mathrm{ro}}$ and, thus, decreases the truncation error of SBEM. Hence, the performance gain by spatial rotation is larger at high SNRs when the truncation error is dominant, but it is not so obvious at the lower SNRs when the noise dominates.

Fig. 11 illustrates the average achievable sum rate (AASR) for the DL data transmission, which is defined as

$$\mathrm{AASR} = \frac{\left(1 - \frac{T_{\mathrm{pilot}}}{T}\right) \sum_{g=1}^{G^{\mathrm{dd}}} R\left(\mathcal{U}_g^{\mathrm{dd}} \middle| \mathcal{P}\right)}{G^{\mathrm{dd}}}$$

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TABLE II
COMPARISON OF THE PROPOSED SBEM WITH EXISTING LOW-RANK MODELS

Challenges \ Approaches	JSDM [20]	Yin [18]	CS [17][21]	SBEM
UL Pilot Decontamination	✓	✓	✓	✓
Reduced DL Training Overhead	✓	✓	✓	✓
No Need for Covariances	×	×	✓	✓
Low Complexity	×	×	×	✓
Reciprocity for FDD Systems	×	×	×	✓
Blind Angle (partial channel) Tracking	×	×	×	✓

where T_{pilot} denotes the length of pilot used for training. We take $T_{\text{pilot}} = G^{\text{dt}}L = 32$ for the proposed SBEM and $T_{\text{pilot}} = M = 128$ for the conventional LS. To make the comparison fair, the overall training power and the overall data power within the coherent time T are set as the same for each method. The DL training procedures for both methods are similar to Fig. 7, with $\tau = 16$ and $L = 32$. Users are scheduled by Algorithm 1. It is shown in Fig. 11 that the AASR of the proposed SBEM is much higher than that of the conventional LS when T is relatively small or when SNR is relatively low. When T becomes large, the training length of LS is small compared with T, and then the AASR from the conventional LS will approach that from the proposed SBEM. Nevertheless, the main disadvantage of the conventional LS method mainly lies in the demand of $L = M$ orthogonal training sequences and the high computational complexity.

Next, we show the bit error rate (BER) performance under quaternary phase-shift keying modulation for the DL data transmission in Fig. 12. Four kinds of CSI are compared, i.e., perfect CSI, CSI from the proposed SBEM method, CSI from the conventional LS method, and CSI from the joint spatial division multiplexing (JSDM) [20]. To keep the comparison fair, the overall training power is set as the same for each method. It is seen that the BER achieved by JSDM is better than that of the proposed method by about 0.5 dB due to its exploitation of real channel covariance matrices. However, JSDM needs $M \times M$ DL channel covariance matrices of all users and needs to perform EVD for all these channel covariance matrices to obtain the exact basis vectors of the channels. On the other hand, the proposed SBEM utilizes constant Fourier basis vectors for all different users, which eliminates the demand for the DL channel covariance matrix, and the selection of the spatial signatures can be achieved by efficient FFT. Moreover, both the low-rank methods, i.e., SBEM and JSDM, perform better than the LS method at the low-SNR region due to the inclusion of less noise power during channel estimation.

Finally, we consider the situation with user mobility. Fig. 13 compares the DL MSE performances of the proposed SBEM, the covariance-based JSDM [20], and the CS-based method in [21], where the instantaneous AS is set as 4° at each time and the statistical AS[9] is set as 4°, 14°, 16°, and 20°, respectively. It can be seen that, as the statistical AS increases, the MSE performances of JSDM obviously deteriorate, whereas the MSE curves from SBEM and the CS-based method [21] are not affected as much. The reason lies in that, as the statistical AS increases, channel covariances will cover AS that is too broad and, thus, are not accurate for instantaneous channel estimation. Instead, SBEM and the CS-based method [21] do not rely on the statistics and thus are more suitable for the mobile user cases. Meanwhile, SBEM has a smaller error floor than the CS-based method for the reason that SBEM offers a sparser channel representation by spatial rotation, whereas the CS-based method does not harness the antenna array theory to enhance the channel sparsity and thus has a relatively larger sparsity level, resulting in lower channel estimation accuracy.

[9]The statistical AS is the total range of AS as user moves and is considered for channel covariance acquisition.

VI. CONCLUSION

In this paper, we have proposed a unified UL/DL channel estimation strategy for TDD/FDD multiuser massive MIMO systems. We exploited the physical characteristics of ULA and proposed a simple DFT-based SBEM based on the array antenna theory. It is shown that the basis vectors for different channels formulate the spatial beams toward the users, and thus, users could be spatially separated during both the training and the data transmission. The conventional headache of pilot contamination is then immediately relieved. Meanwhile, the UL spatial signatures could be also used for DL training based on the angle reciprocity of electromagnetic propagation, making the proposed SBEM applicable for both TDD and FDD massive MIMO systems. Channel estimation algorithms for both UL and DL were carried out with very few training, and the amount of the feedback could be significantly reduced. To enhance the spectral efficiency, we also proposed a user scheduling algorithm during the data transmission period, where the spatial signatures of users were exploited again. Compared with existing low-rank models, the proposed SBEM offers the advantages such as low estimation complexity, low overhead requirement, and applicability for both TDD and FDD systems. The detailed comparison of SBEM with existing low-rank models is shown in Table II. Various numerical results were provided to demonstrate the effectiveness and the superiority of the proposed method.

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